Research Proposal:
Rare-Stratum Fidelity in Privacy-Preserving
Synthetic Clinical Data

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Introduction

This project develops synthetic clinical data that preserve the statistical properties of real data while protecting patient privacy. These two goals conflict most sharply inside rare strata: uncommon diagnoses and small comorbidity groups carry little weight in global fidelity metrics, yet they are often what clinical research cares about most, and where a generator is most likely to lose structure or memorize individuals. In an earlier study of prior sensitivity in Bayesian logistic regression (Bayesian Prior Sensitivity), I found that estimate stability was driven by a predictor's positive-case count, not sample size. This project examines whether synthetic fidelity behaves the same way at the stratum level.

Methods

As a starting point, I would begin from the data and move up to the tree-based generator, holding the generator fixed and varying only the input so that any change in fidelity is attributable to the input.

1. **Sweep the positive count**: Hold everything fixed, including sample size, and vary only the rare stratum's positive count. Compare the structure recovered from the real subset against the CART-synthesized data (`synthpop`) at each count.
2. **Sweep the sample size**: Hold the positive count fixed and vary the sample size instead, to separate the count effect from sample size and prevalence.

I will first validate this on a simulation with a known ground truth, then extend it to the tree-based generation on MIMIC-IV. All work will be conducted in R using `synthpop`, `rstanarm`, and `ggplot2`.

Goals

- Understand how fidelity and privacy are balanced inside rare strata, where preserving structure and protecting individuals pull in opposite directions. This tradeoff is the part I most want to study.
- Compare the two experimental levers directly, sweeping the positive count against sweeping the sample size, to see which one actually drives rare-stratum fidelity.
- Connect this to the tree-based generation used in the project, moving from the input data up to the CART-based synthesizer and examining where it succeeds or fails on rare strata.

Preliminary Result

As a first step, I ran the simulation with a known interaction (`chf:lactate`, true value 1.5) in a rare stratum, sweeping its positive count with sample size fixed at 500 (single draw). The real-data estimate stabilizes near the truth by a count of about 20; the synthesized estimate does not recover until roughly 80.

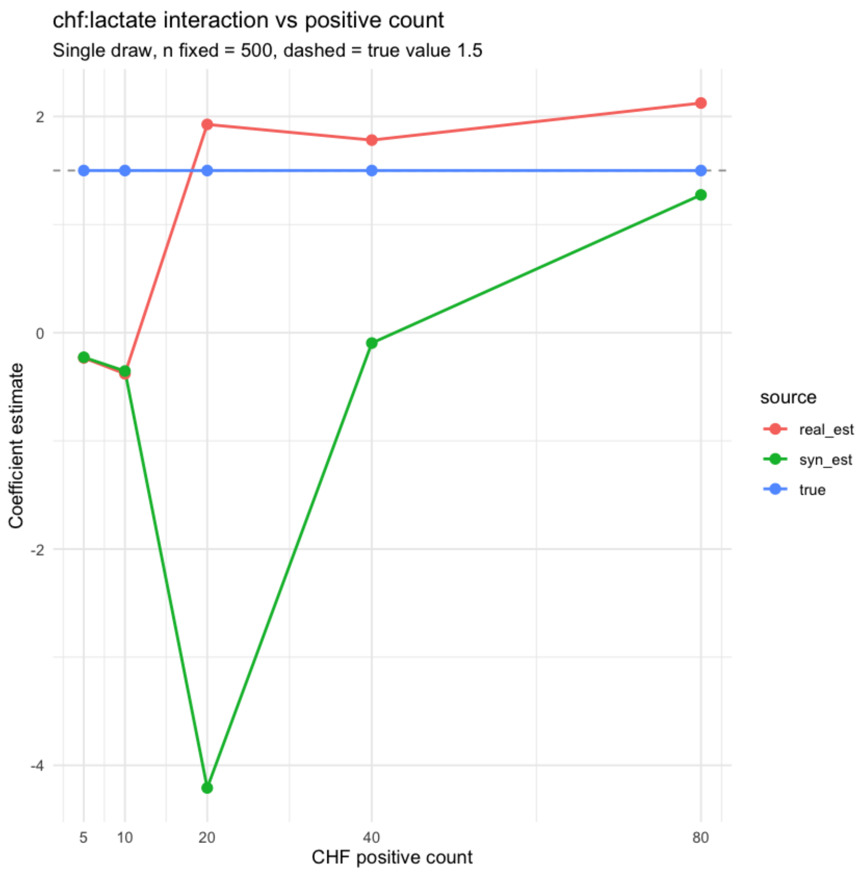

Figure 1: Recovered `chf:lactate` interaction versus rare-stratum positive count, sample size fixed at 500, single draw.